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The search for topics related to electric mobility: a comparative analysis of some of the most widely used methods in the literature

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Abstract
Identifying the topics addressed in a corpus is one of the primary concerns of automated text analysis. This paper aims to contribute to the comparative analysis of various methodologies. Specifically, a comparison is made of the results obtained by applying the most prevalent topic identification techniques to the same corpus. The analysis is conducted on a large database of original text created from an e-mobility newsletter. To evaluate the outcomes of the methodologies, two criteria are used. First, the semantic coherence and similarities of the various methods are assessed. The second step involves processing the degree of association between the topics identified by the various models.

Keywords Topic detection · Text mining · Cramer's V · Coherence indexes · Semantic similarities · Electric mobility

1 Introduction

The continuous proliferation and availability of digitised textual information has increased the demand in academia and industry for systems and algorithms capable of extracting information of interest from unstructured, semi-structured and fully structured textual data, particularly over the last two decades. This data availability makes it possible, on the one hand, to conduct qualitative analyses of document collections in all research contexts and, on the other, to develop Apps with the most diversified goals in the context of ordinary life. Many Text Mining approaches [1, 8, 9, 27, 66] effectively combine linguistic resources, computational methods, and statistical techniques for the analysis of texts, representing a highly interdisciplinary field that assists researchers from various disciplines (such as social science, psychology and education) in utilising and interpreting the results of text analysis of large corpora. In general, when applied to scientific research, these processes do not only involve the training of models, but also require numerous additional procedures, such as the

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pre-processing of texts and the transformation and reduction of the dimensionality of the data to be analysed.

Among the many objectives that can be defined within a text analysis, the clustering techniques of documents, or document fragments, based on their content similarity and the identification of the topics covered in the collection of documents are of particular importance [2, 16, 19]. A clustering phase applied to textual data enables the formation of groupings of texts that share similar terms. The phase of terms-based clustering is an unsupervised classification that reflects the semantic similarity between texts. This conceptual homogeneity makes explicit the theme or semantic trait prevalent in each group of texts, which can be summed up in a category not defined a priori but obtained by reading the characteristic language (terms) of each group of texts.

In recent years, numerous methods for identifying topics in a corpus have been developed, the most prominent of which can be grouped into three main categories: probabilistic methods, matrix factorisations and clustering techniques. The first group of methods comprises the Probabilistic Latent Semantic Analysis (PLSA) [34–36] and one of the most frequently used methods in this context which, starting from a first work by Blei et al. [11], was developed by Griffiths et al. [31], who introduced the Latent Dirichlet Allocation (LDA), a generative model for identifying topics within a corpus. The second group of topic detection methods contains the factorial-based techniques. These seek to reduce the number of terms necessary to characterise a collection of documents by representing the most pertinent information in a low rank vector space. The dimensionality reduction is achieved by considering linear combinations of the original terms expressing the content of the documents and visualising these latent association structures onto factorial maps. This group encompasses the first attempt to identify unobserved "latent factors" namely, the Rotated Factor Analysis (RFA) [67], a method used to identify and interpret "latent factors" within a set of observed data. These factors are represented by linear combinations of observed variables and are rotated to facilitate the interpretation of results. Following that, the most popular methods in the literature are the Latent Semantic Analysis (LSA) [22], the Lexical Correspondence Analysis (to select the most relevant factors that explain variability within the corpus, used in conjunction with a hierarchical cluster analysis) (LCA) [44], the Non-Negative Matrix Factorisation (NMF) [33] and the Logarithmic Analysis (ALO) [39], which "produces results similar to those of LCA, with less sensitivity towards outliers as a consequence of the logarithmic shrinkage" [43, p. 218].

In the clustering-based methods, each algorithm assumes that the textual units included in the same cluster discuss the same topic. For each cluster discovered, the centroid is used as a representative, with the top ten words (in terms of chi-squared test-values or Term Frequency Inverse Document Frequency, TF-IDF, weights) serving as the topic's keywords. These methods include DBSCAN [25] (Density-Based Spatial Clustering of Applications with Noise), Spectral Clustering [46] and all those based on K-means, such as Sequential K-means [37], Spherical K-means [4], Kernel K-means [63]. One of the most widespread clustering methods applied in text analysis in sociology and psychology is the Reinert method [50, 55], also known as the ALCESTE method (ALC) [52], which is the name of the software that first implemented this type of method.

Lebart [43] focuses on the comparison of expert classification and various techniques (LDA, RFA, ALO, LSA, and NMF) applied to a given medium-sized and homogeneous corpus, i.e. Shakespeare's 154 Sonnets, in order to compare some characteristic features of different methods in a concise fashion. In his analysis, Lebart concludes that the various methods "concur on the same topics […] despite the amazing variety of their theoretical backgrounds". He also emphasises that the results depend on "various parameters and options"

and that "exploratory or descriptive tools… have been essential to visualize the complexity of the process and to assess the obtained results" [43, p. 222].

The purpose of this paper is to contribute to the broad debate on methods for text analysis, as summarised by Lebart [43], but by comparing a different group of methods than the ones he proposed and implementing analytical tools to visualise and assess the comparison among methods. To this end, a large original textual database created from a newsletter on environmentally sustainable mobility issued daily by electrive.com is used. The objective is to identify the socio-technical domains as well as agents and artefacts associated with each domain in an emerging market system undergoing rapid transformation. Russo et al. [59] have already analysed this data set (consisting of news items published between 21st August 2018 and 15th September 2021), focusing on a topic detection based on Lexical Correspondence and cluster analysis in order to identify socio-technical domains and provide a complete entities disambiguation with respect to agents and artefacts mentioned in the news items. This paper updates the database (with news items through 8th March 2022) and explores some topic identification methods. Each news item focuses primarily on one topic (for example, the production of an electric bus model), although other topics may be mentioned (for instance, the batteries used). The size of the corpus is appropriate for comparing information retrieval and textual unit classification methods.

In the paper, reference is made to expert classification (directly provided by the newsletter editors) and to five alternative methods for clustering/topic detection, namely, LDA, LSA, LCA, NMF and ALC. With regard to the selection proposed by Lebart, the RFA and ALO methods have been excluded while maintaining the LCA and ALC methods in the comparison, which are more widespread in the literature, especially in the fields of sociology and psychology. The rationale behind the choice of these methods is given in the paper, in addition to a discussion of the characteristics of the various methods in relation to the corpus under analysis and the research questions on which the paper focuses. The diverse range of methods available in the literature for identifying topics underscores the fact that no one method is more appropriate than another for any corpus and any research question. In the cross-methods comparison, in which a given corpus (a newsletter on electric mobility) is used to identify emergent agents and artefacts in the various domains of the electric mobility eco-system, a preliminary research question is precisely: what are the various domains of this eco-system? The objective of disentangling these domains as subsystems is to identify topics that define them at increasing levels of aggregation, followed by the association of recognised entities (agents and artefacts) for each domain-topic in the eco-system. A significant result of the analysis is the association of text units to the various domains-topics, which enables the association of agents and artefacts occurring in the corpus of various news items to the topic emphasised in each group.

Automatic analysis enables speed, consistency, and reproducibility, and generates a systematic comparative and contextual analysis, thereby overcoming the limitations of classifications and analyses of large corpora based on the subjective opinion of the individual who reads and classifies each text one by one. On the one hand, the limits deriving from the expert reading of large quantities of texts are often overlooked, despite the fact that there can be significant distortions which impact the interpretation of the results. On the other, the adoption of automatic techniques for the topic detection/clustering process of text analysis must be characterised by transparency in the specification of the methods of analysis and in the interpretation of the results, thereby favouring their reproducibility both to qualify their scientific character and to facilitate their systematic use over time, or for corpora with similar characteristics. In light of these considerations, the paper identifies a number of crucial challenges that must be addressed explicitly when employing topic models.

The paper presents data (Sect. 2), methodologies (Sect. 3) and results (Sect. 4), with an emphasis on identifying an optimal number of topics, comparing the relative semantic coherence and semantic similarity of topics in each method and, finally, examining the similarity of topics across methods. Discussion of results (Sect. 5) pertains to the implementation of various topic detection and textual unit classification models. The conclusion (Sect. 6) provides a summary of the results of the cross-method comparison and their implications for addressing the research questions on socio-technical domains in sustainable mobility.

2 Data

The data under examination is a collection of English-language news items published by electrive.com, a daily online newsletter offered by Rabbit publication GmbH. It covers a wide range of significant information on developments in electric transport in Europe, the USA and China. The data source is also relayed on the main social media, including Twitter. Using the Twitter API, tweets from 12th September, 2018 to 8th March 2022 were downloaded from the timeline of the electrive.com Twitter page. The link to the news URL was identified within each tweet. Using a web data extraction procedure (web scraping), the title, entire text, associated tags, category, publication date and links to the information sources were extracted from the news page for every piece of news. The electrive.com corpus is composed of 5216 news items (title and full text) published in the period 12th September 2018 to 8th March 2022 and consists of a vocabulary of 54,230 different words (i.e. types) totalling of 2,175,691 word occurrences (tokens). The smallest news item in the corpus is composed of 35 terms, while the largest is composed of 3858, with an average of 417 terms per news item (8% of news items have less than 200 terms, 47% have between 200 and 400 terms, 38% have between 400 and 700 terms and 7% have more than 700 terms).

Of the ten categories proposed by electrive.com, namely, Air, Automobile, Battery & Fuel Cell, Energy & Infrastructure, Fleets, Politics, Short Circuit, Two-Wheeler, Utility Vehicles, Water, the main one, “Automobile”, encompasses nearly 38% of the news, followed by “Battery and fuel cells” and “Energy & infrastructure”, each with nearly 14% of the news items. This expert categorisation produced a large domain, “Automobile”, that includes a range of different technologies, specific features of the electric engine, business models and policy interventions. In order to provide a sociotechnical analysis of the present transition in the mobility sector, it is necessary to identify agents and artefacts in the various electric mobility subsystems. In what follows below, this result is achieved through the application of a number of automatic topic detection models and the classification of textual units under analysis (the news items in this corpus).

3 Methodology

Automatic analysis of textual data makes it possible to extract information of interest from a collection of texts, or “corpus”, which in this case is composed of a collection of news items. Each piece of news (which represents the textual unit of analysis) corresponds to a text describing an event involving electric vehicles. A text is therefore composed of a series of sentences which in turn are composed of a number of words, used to describe the narrative fact. Each word consists of a sequence of characters between two separators and represents the elementary lexical unit. Focusing on the topic detection and textual unit classification,

for all the methods under comparison in this paper the same method of pre-processing text (Sect. 3.1) and the same vector space model representation of the news item and the selected elementary lexical unit under analysis (Sect. 3.2) are adopted. The five topic detection models and the cross methods comparison are also briefly described (Sects. 3.3 and 3.4, respectively).

3.1 Pre-processing of the text

Words have different functions within a sentence. Some have a language structure function (articles, prepositions, conjunctions, etc.) and some a semantic content. Among the latter can be seen terms that describe and qualify objects and events (common nouns and adjectives), others that indicate actions (verbs), others again that indicate places (toponyms, and, finally, those that indicate proper nouns. Pre-processing of the text is, therefore, a necessary operation to ensure that only the most suitable content words are selected to highlight the topics in the corpus under examination. Based on that aim, a text mining strategy was defined. The first step facilitated an analysis of the texts consisting in structuring the textual information in a lexical and textual database. This was carried out using TaLTaC2[1] software [12, 13]. Due to the specific characteristics of news items clearly and carefully written, and a grammatical tagging of the vocabulary words, it was possible to distinguish easily between common nouns (content terms with lowercase initial) and proper nouns (non-structure terms with capital initials), thus identifying 522,916 words, 4,439 proper nouns[2] (of people and companies), 1,626 acronyms[3] (defined by all capital letters). Using Taltac resources, all the types (simple and compound) referring to nations (including national adjectives), which totalled 209 terms mentioned in the text, were identified. As already noted, grammatical tagging made it possible to distinguish between the different grammatical types of words (structure words versus content words) and also to lemmatise them, that is, to relate each word to its basic form from which an entire nominal, adjectival and verbal inflection system is derived, thus reducing the forms under analysis.

Grammatical annotation of the terms that compose a vocabulary is quite a complex process for which, generally speaking, there are two possible alternatives: "out-of-context" grammatical tagging and "in-context" grammatical tagging. The former enabled the grammatical annotation of the words of the vocabulary using linguistic resources, following which a comparison is made between vocabulary words and lists of grammatical terms. This method of annotation gives, as output, a word which is grammatically noted in a very precise manner (absence of false positives), but with a large number of ambiguous annotations (presence of false negatives), triggered by the different grammatical classes which can be associated with the same word. The latter method consists of formulating algorithms which analyse the forms in the context in which they occur. The output for this method is a completely annotated vocabulary (absence of false negatives) but inevitably a certain number of forms

[1] TaLTaC is the acronym for "Trattamento automatico Lessico-Testuale per l'analisi del Contenuto" (lexical-textual automatic treatment for content analysis). It was designed for automatic text analysis in the dual logic of lexical-textual analysis and text mining, https://www.taltac.com/ (visited on 23rd July 2023) Taltac has been developed in Italy since the 1990s by Sergio Bolasco and his research team, building in the context of the French school of textual data analysis and disseminated in research published in JADT conferences (*Journées d'Analyse Statistique des Données Textuelles*) – with Tullio De Mauro's contribution on linguistic resources, https://www.taltac.com/presentazione/ (visited on 23rd July 2023).

[2] Among these the five most frequent are (frequencies in brackets): *Tesla* (4,217), *Volkswagen* (1,752), *Audi* (1,160), *Hyundai* (1,076), *Ford* (1,073).

[3] Among these the five most frequent are (frequencies in brackets): *EV* (2,112), *CEO* (1,860), *BMW* (1,764), *VW* (1,485), *SUV* (1,096).

will be noted erroneously (presence of false positives). This pre-treatment stage is strategic for the subsequent development of the analysis, based in the comparison of topic identification methods. In short, it enables the selection of only those terms of semantic importance, excluding not just forms of language structure but also proper names (recognisable by the use of capital letters). Although both of these methods are valid, in what follows out-of-context grammatical tagging, implemented with Taltac, will be used. Finally, implementation of a lexical-textual model [14, 48, 49] enabled the recognition of multiword expressions present in the texts. This operation yielded lexical analysis with less semantic ambiguity, since their recognition facilitates the identification of both idiomatic MWEs and collocations [64] and, more generally, the definition of the specialised technical terminology of a corpus.

In order to detect the topics in the subsequent stages of the analysis with which to construct a vector space model for subsequent analysis stages, only common nouns and adjectives were selected for each news item (for a total of 8,489 terms), such as lowercase common nouns and adjectives (5,871 simple words and 2,618 multiword expressions).

3.2 Vector space model

A vector space model representation [60] was generated, by representing each news item as a vector $\mathbf{d_i}$ in the space spanned by the q words, the selected terms.

$$\mathrm{D}_i = \left(W_{i1}, \ldots, W_{ij}, \ldots, W_{iq}\right)$$

where *Wij* represents the term frequency of the jth word in the text-vector.

For all the methods discussed, the same vector space model was chosen, i.e. the vectorisation of the textual data into a matrix of the type $\langle newsitems \times selectedterms \rangle$, which does not take into account the possibility of reducing the news items into smaller, homogeneous fragments. It should be noted, however, that, although there are different sizes in the news texts, each one stands for a news item. The initial assumption, therefore, is that they define a single semantic item. Specifically, a document-term matrix D $\langle newsitems \times selectedterms \rangle$ (5216 × 8489) was defined, to be analysed using the different methods selected in order to identify the topics covered within the corpus. In addition to expert classification, five methods for clustering/topic detection were implemented in the analysis of the corpus: LDA, LSA, NMF, LCA and ALC. The following libraries have been implemented in R: *topicmodels* (for LDA), *LSA, NMF, FactoMiner* (for correspondence analysis), *quanteda* and *rainette* (for preparing the dataset and processing the Reinert method, respectively).

3.3 Topic detection methods

3.3.1 Latent Dirichlet Allocation (LDA) [11, 31]

LDA is a generative model used for the analysis of topic models. It entails a probabilistic approach in which each document is considered as a combination of topics, each of which, in turn, is a distribution of terms. Specifically, the model assumes that the distribution of topics in documents and of terms within each topic always follow a Dirichlet distribution.

The goal of LDA is to identify the probability distributions of topics and their associated terms that best explain the corpus of texts. The inference step aims to find the parameters of the process that are most likely to have generated the corpus. In this context, LDA employs Bayesian inference to estimate the model's parameters, in particular the probability distributions of topics and terms. In practice, various computational algorithms can be used in LDA

to estimate the parameters of the model and infer the topics in the corpus. This is because the "a posteriori distribution", which represents the conditional distribution of the topic structure given the observed documents, is not computationally tractable. One of the most commonly used computational algorithms for LDA is Gibbs sampling. The results of the inference process lead to the definition of a set of distributions describing the probability of a particular topic for each specific document in the corpus and a set of distributions associating the probability of each term to belong to a topic. The meaning of the topics can then be derived by interpreting the terms sorted in descending order of probability.

3.3.2 Latent Semantic Analysis (LSA) [22]

LSA is a textual content analysis technique that uses a dimensionality reduction approach based on the factorial method. Starting from a vector representation of the semantic space, it uses a Singular Value Decomposition (SVD) to reduce the dimensionality of the co-occurrence matrix of terms and identify latent concepts within the text. Generally speaking, SVD decomposition is combined with the use of a TF-IDF weighting of the co-occurrence matrix which results in a vector representation of the corpus that takes into account both the frequency and importance of terms within the corpus. SVD consists of the decomposition of a term co-occurrence matrix D into three components:

$$\mathrm{D} = \mathrm{U} * \mathrm{S} * \mathrm{V}^{\mathrm{T}}$$

where U is an orthogonal matrix m $\times$ k ⟨documents $\times$ terms⟩, V is an orthogonal matrix n $\times$ k ⟨terms $\times$ topics⟩ and S is a diagonal matrix k $\times$ k ⟨topics $\times$ topics⟩, with non-negative elements in descending order along the diagonal, where k is the desired number of topics, set a priori. The U matrix represents the relationship between documents and topics, while the V matrix represents that between terms and topics. The S matrix indicates the relative importance of the topics.

Dimensionality is reduced by selecting the first k columns of U, S and V. This operation corresponds to the projection of the original matrices into a k-dimensional space, where k is generally much less than the number of terms in the corpus. The SVD thus enables the term co-occurrence matrix to be represented in a k-dimensional semantic space.

Axis rotation is a process that is applied to the matrix of latent components extracted via LSA to improve interpretability. The objective in this case is to find a new representation of the latent components in which each term is associated with only one component and each component with a coherent set of terms. This may be useful for identifying the most significant semantic relationships between terms and for enhancing the interpretation of the results.

When applied to LSA, the most commonly used rotation method is Varimax. This consists of maximising the variance of each individual LSA factor, such that the weights of the terms associated with a single topic are more concentrated on a few terms rather than being evenly distributed. This axis rotation technique thus makes it easier to analyse the document and term vectors. Interpretation of the topics is then done by selecting those terms that make a greater contribution to the factorial axes in the rotated matrix of V.

3.3.3 Non-negative Matrix Factorization (NMF) [33]

NMF is a factorial dimensionality reduction technique, but unlike SVD it requires the constraint that the matrices be non-negative. This restriction has the advantage of obtaining more

easily interpretable results, since the entities that are extracted from the matrix are always positive. In particular, the NMF, applied to a co-occurrence matrix, aims to find two non-negative matrices, namely, a topic matrix (describing the distribution of topics within the corpus) and a terms matrix (describing the distribution of words within each topic):

$$\mathrm{D} \approx \mathrm{W} * \mathrm{H}$$

where D is a $m \times n$ ⟨documents × terms⟩ matrix, W is a m × k ⟨documents × topics⟩ matrix (distribution of the topics among the documents) and H is a k × n ⟨topic × term matrix⟩ (significance of the terms among the topics), while k is the desired number of topics, set a priori.

In NMF, one of the most commonly used objective functions is the Kullback–Leibler divergence, which measures the discrepancy between the input data and the reduced rank approximation. Specifically, the objective of the NMF is to find the W and H matrices that minimise the KL divergence between the original matrix and the reconstructed matrix obtained by multiplying the W and H matrices. The NMF also works on TF-IDF-transformed data to assess the importance of a term in a set of documents. The semantic meaning of the topics can be derived from the terms that best characterise the factorial axes of the H matrix, while the topics that characterise the documents will be those with the largest coefficients in the columns of the W matrix.

3.3.4 Lexical Correspondence Analysis (LCA) and cluster analysis [44]

The main objective of LCA is to uncover the associations and patterns of co-occurrence between words in different textual contexts. It aims to provide insights into the semantic relationships and similarities among words based on their patterns of occurrence. LCA builds upon the traditional Correspondence Analysis (CA) technique [6, 7, 30], which is commonly used for exploring relationships between categorical variables. However, instead of analysing categorical variables, LCA focuses specifically on words or lexical units. The process of LCA involves constructing a contingency table that represents the co-occurrence frequencies of words in different contexts or documents [5]. This document-term matrix serves as the input for the analysis. LCA then calculates the chi-square distances between the rows and columns of the document-term matrix, which measure the dissimilarity or similarity between documents and between words.

Subsequently, LCA performs a dimensionality reduction technique, typically Singular Value Decomposition (SVD) [29], on the chi-square distances to extract the underlying latent dimensions or factors which capture the semantic relationships and associations between documents based on co-occurrences of words. Once the dimensions have been obtained, LCA can generate graphical representations, such as the factorial map, to visualise the positions of documents or words in the semantic space. Documents that are closer to each other in the map are more likely to have similar word co-occurrence patterns, thus reflecting the semantic similarity between documents.

By applying cluster analysis to the dimensions obtained from LCA, it becomes possible to identify groups of documents based on semantic associations. To avoid excessive sensitivity towards shorter texts, these can be considered as supplementary elements in the correspondence analysis. They do not affect the definition of the factorial subspace, but they can still be represented on the factorial axis according to their similarity to the groups defined through the active elements. For more details see Greenacre [30, p. 92]. The terms that characterise each document group represent their main topic and are identified through the calculation of

lexical specificity [41]. This statistical measure is based on the hypergeometric distribution, which efficiently highlights a set of relevant terms from a subset and is utilised in various Natural Language Processing (NLP) applications [10, 16, 19, 24]. The state of the art in clustering is provided by a huge literature (revised systematically in [37]), developed in a variety of scientific fields with different languages and focusing on the most diverse problems such as clustering heterogeneous data, definition of parameters and initialisations and the threshold in hierarchical clustering, as well as the problem of defining the optimal number of groups. Thus hundreds of techniques and algorithms can be found the literature, each of which attempts to overcome some of the limitations of the previous methods. While this paper does not go into the impossible challenge in defining what the best classification method might be, a hierarchical agglomerative clustering based on Ward's aggregation method [68], with Euclidean distance, is applied to the coordinates of the texts on the first 10 factors obtained through the Correspondence Analysis. Again, the choice of the number of factors to be used for cluster analysis is the result of a common sense choice. Each factor represents a semantic aspect on which the documents are polarised. Performing a cluster analysis on the basis of the first 2 factors will result in 3–4 groups of documents based on the combinations of the positive and negative polarisations of each factor. Using an excessive number of factors on which to cluster documents may generate excessively detailed information on a given semantic aspect.

3.3.5 Reinert method, ALCESTE method (ALC) [53, 54, 56]

This method is based on a top-down hierarchical classification (divisive partition) of elementary context units, that is, the news items in the database. Given the relatively small size of each document in the corpus, fragmentation of texts is not applied. The algorithm described by Reinert [53, 54, 56] is based on a series of bipartitions constructed so that each maximises the chi-squared value of the groups of elementary context units and a transformation of the matrix D into a Boolean table (absence/presence) that cross-references the elementary context units with the content words in the corpus [20, 50]. The descending hierarchical procedure considers as results only those classes of context units that are associated twice in the same cluster, starting from different a priori groupings of context units (basically by varying the number of words that make up each context unit). Thus, the modelling does not capture all the available information, but only that which has a high degree of internal coherence, proving that it only represents lexical worlds (i.e. those stable systems of semantic representation) that take shape in relation to each other as they are the most diverse. Once the text groups have been defined, the terms defining each topic are identified through their positive characterisation calculated in terms of the chi-squared test. The expert interpretation of those terms is then used to label the various topics.

3.4 Cross method comparison: criteria and tools

The criteria and tools that will be used in the cross-method comparison are as follows: (a) the identification of the number of topics to be compared for each topic detection method; (b) the definition of criteria to check their semantic coherence; (c) the implementation of tools to compare the results between methods, i.e. the similarity of topics between methods; and (d) the implementation of tools with which to compare the classification of documents.

3.4.1 Number of topics to be compared

For each method of topic detection, the optimal number of topics, if any, depends on various factors, such as the size of the corpus, the objective of the analysis, the complexity of the texts or the method used. The objective of the analysis is to categorise documents according to their primary topic. The corpus analysed is comprised of news items, hence not particularly lengthy texts, and can be considered to be topic-focused. With the exception of LCA, the a priori choice of the number of topics is a constraint that characterises all the methods used. In order to compare the different methods, it would be ideal for this analysis to have an equal number of topics from the different methods. To do this, the number of topics defined by the experts as a benchmark could have used. However, as already mentioned in the data description, the ten categories proposed by the experts include three very general and large topics that cover 66% of the news items: "Automobile" (38%), "Battery and fuel cells" (14%) and "Energy & infrastructure" (14%). It was therefore mandatory to find a number of topics greater than 10 that disaggregates the largest classes (in particular).

With regard to the number of topics, some general considerations apply. As the aim is to identify the main topics, a very high number of topics could lead to an overlapping thereof in the individual documents, which in turn would make it difficult to classify them unambiguously. Conversely, if a small number of topics is chosen, the definition of the topics would be too general and classification of the documents would be too rough. Based on methods proposed in the literature, several criteria have been applied and discussed in detail in the results Sect. 4.1.

3.4.2 Semantic coherence

Following the literature, [45, 57, 65], a check for the semantic coherence in topic models is checked by applying two measures of coherence. These are the UMass score, based on a log-conditional-probability measure, and a variant of the UCI metric, based on normalized pointwise mutual information. Both are intrinsic measures based on the co-occurrences in the corpus of a given number of words defined in each topic.[4]

Then, in the cross-method comparison, two parameters have to be defined: the number of topics and the number of terms per topic. The results of the coherence of the topics, as their number varies, are shown in Fig. 1 with respect to two measures of coherence: Mean Log Ration and Mean Normalised Pointwise Mutual Information. It can be observed that there is no optimal number of topics valid for all the methods analysed.

Assuming that a topic is represented by the first N words with the highest probability of belonging to that particular topic, the coherence score measures how similar these words are to each other. All methods define a ranking of the terms associated with each topic. So, for example, LDA defines the probability of a term belonging to a topic, while LSA assigns the factorial score of the term on the axis defining a topic. For a comparison between topics an arbitrary threshold could then be set to select the most important terms (for instance, in LDA those with a probability above a certain threshold). However, this would lead to topics being defined by a different number of terms, making the comparison difficult. In order to make it homogeneous, the same number of terms per topic can be chosen instead. In this case,

[4] The analysis in R uses the *text2vec* library. The selection of the terms identifying the topics is specific to the different methods used, respectively: test-value for LCA; a chi-square test for ALC; the terms with the highest loadings for the LSA [26]; the terms with the highest probability for LDA, the terms with the highest factorization weight in the term-topics output matrix for NFM.

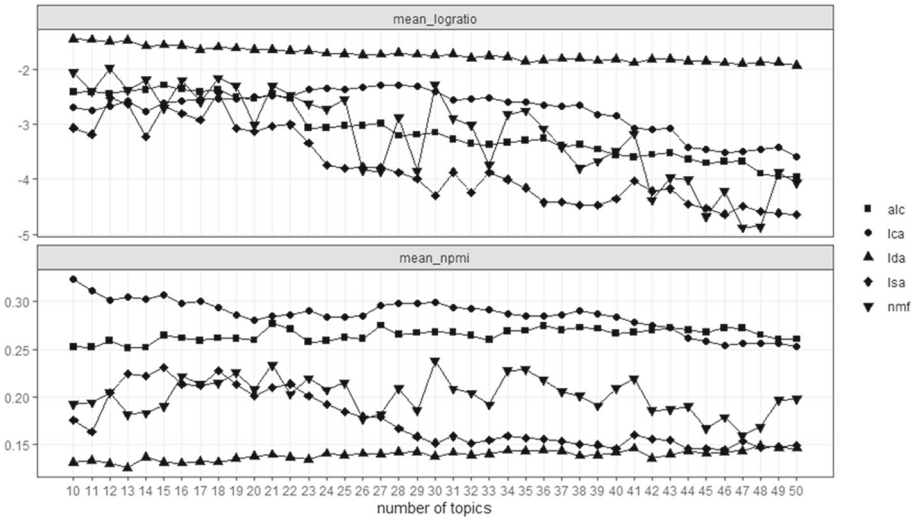

Fig. 1 Coherence of topics resulting from ALC, LSA LDA, LSA, NMF, by number of topics (mean_logratio and mean Normalized Pointwise Mutual Information (mean_npmi))

however, it has to be decided how many terms to choose. For this purpose, it is possible to see, for each method, which number of terms makes the topics most different in the vector space. By examining the average distance between topics as the number of terms varies (using a cosine metric), Fig. 2 below illustrates that, with regard to all methods, it tends to decrease as the number of terms increases. Nevertheless, it is evident that subjects defined by a limited number of terms may be uninformative and difficult to interpret semantically. In contrast, topics defined by a multitude of terms may be less semantically distinct.

From the analysis of the results in Fig. 2, and considering the different trends of the average distance between topics when varying the number of terms, in order to conduct a comparative

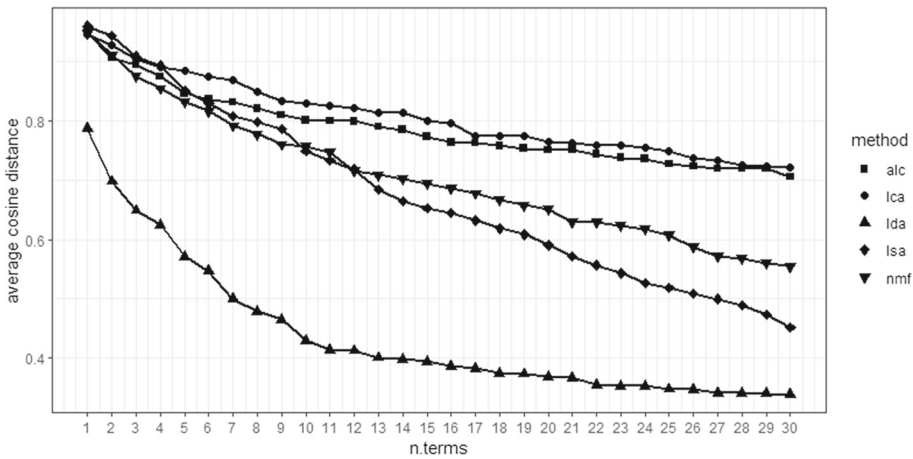

Fig. 2 Average topic distance per method and number of terms

analysis of the topic coherence of the various methods examined, 10 terms were chosen. The results are discussed in Sect. 4.2.

3.4.3 Semantic similarity

Having defined the benchmark of the number of topics for the cross-method comparison, and having assessed the homogeneity of the different topics detected with the methods under examination, the following question arises: to what extent do the different topics look similar to each other across the different methods? The semantic similarity of the results was checked by performing a cluster analysis on the matrix composed by the topics obtained by the different methods and the ten most characteristic words of each topic. A cluster heat-map has been drawn up on this matrix in order to compare the results obtained by the different methods and the degree of association between the topics. Cramer's V was also used to measure the strength of association between the classifications produced by the different methods. The results are discussed in Sect. 4.3.

3.4.4 Classification of documents

In the expert classification, the ALC and LCA methods only associate each document with one cluster. With the other methods, each document is associated with the most relevant topic among the ones identified (for LDA, the topic with the highest probability and for LSA and NMF, the one with the highest value in the factorial plan). With respect to document classification, a metric has been adopted to compare the results obtained with five topic models under analysis and by expert classification. Specifically, heat-maps have been used to visualise the results obtained by pairs of methods in grouping news items. Pairs of methodologies were centred on Cramer's V as an overall indication of the association's strength in order to provide an estimate of the resulting overlap. The results are discussed in Sect. 4.4.

4 Results

In this section, the results of topic identification from the matrix defined during corpus pre-processing and consisting of 5,125 news items and 8,489 content terms are discussed. For each method of topic detection, first of all the results obtained in defining the optimal number of topics are discussed and the benchmark for cross-method comparison (Sect. 4.1) is set. Subsequently, the semantic coherence of the topics obtained in each method (Sect. 4.2) and the semantic similarity of topics across methods (Sect. 4.3) are presented. Finally, there is a focus on the results of the degree of association of classification of textual units according to the expert classification (10 categories) and the five topic models under analysis (Sect. 4.4).

4.1 Definition of the number of topics

4.1.1 Lexical Correspondence Analysis (LCA) with cluster analysis

For the topic classification obtained with the LCA method, several alternative criteria can be adopted in identifying the optimal number of clusters. Using the R Package *NbClust* [18] it was possible to compare 15 different indices (see Annex 1 for details). In what follows, it was necessary to discard solutions with less than 8 clusters, since this figure is less than the expert

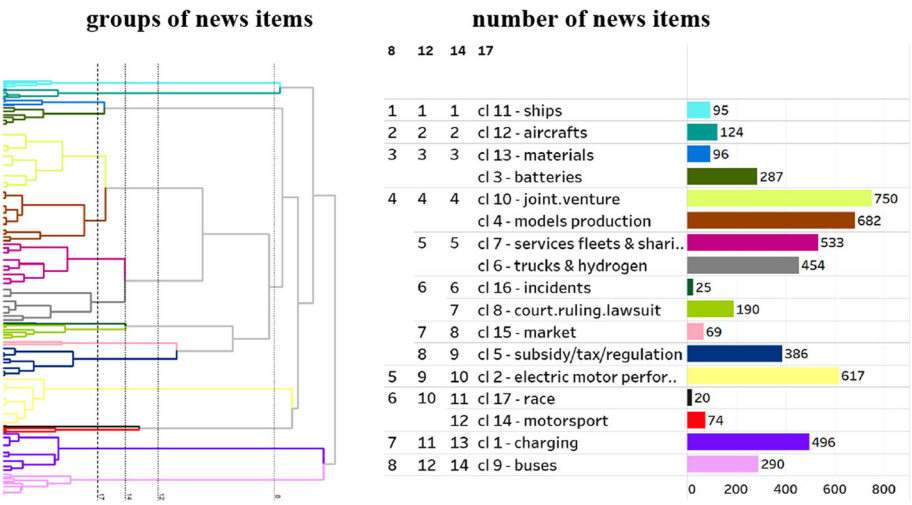

Fig. 3 Dendrogram: results of the LCA method, with 8, 12, 14 and 17 groups

classification at 10 categories. Among the results, it can be noted that both Scott's [62] and Ratkowsky's [51] indices return a maximum at eight clusters. For the Calinski-Harabasz [15], Hartigan [32], Krzanowski and Lai [40] and Rubin [28] indices, the maximum is 12 while according to the Davies and Bouldin [21] and to the Silhouette [58] it is 14. Finally, according to the CCC index [61] the number is approaching the maximum when it is at selections above 12 clusters. Figure 3 shows the dendrogram of the LCA method, with a hierarchical clustering applied on the first 10 factors of the correspondence analysis. Lines refer to the cut-offs at 8, 12 and 14 clusters obtained according to different methods. An additional cut-off at 17 clusters has been defined by interpreting the results of further branches of the dendrogram.

The interpretation of each cluster at different levels of aggregation in the hierarchical branching was carried out by observing the characteristic terms of each cluster, sorted in descending order of test-value [41, 44]. In what follows, the clusters resulting from the progressive cut-offs are discussed, including with respect to the expert classification (the cross tabulation presented in Table 1). The 8-cluster cut-off defines a minimal disaggregation of topics that is more aggregate than the expert classification. This means that is not satisfactory for the purposes of the subdomains analysis of this paper. With the 12-cluster cut-off, the large cluster 4 that groups two thirds of news items regarding the category "Automobile" is broken down into five groups. One of them is still very large (with 27.6% of news items). A second group, which accounts for almost 19% of news items, deals with other-than personal car mobility. Three smaller groups detail, respectively, incident, market issues (demand/supply, prices) and, finally, subsidies and regulation. With the 14-cluster cut-off, two small clusters are decomposed. One refers to "incidents" and "court ruling lawsuit", while the other comprises two news groups, namely, race and motorsport. Aiming at more specific and homogeneous socio-technical domains, the evaluation examined the results of further cut-offs, reaching significant results with one at 17. The first result refers to cl-3 on batteries. It remains combined with all the previous cut-offs (8, 12, and 14, but also at 15 and 16), while the 17-cluster cut-off splits two important groups of news items. One of these, on battery production techniques, is separated from those on materials used in battery production, an issue that is extremely significant in the analysis of the agents involved and the geopolitics of the emerging market

Table 1 Expert categories vs LCA, at 12 and 17 clusters

8	12	14	17	Automobile	Battery&FuelCell	Energy&Infrastructure	Air	Fleets	Politics	ShortCircuit	Two-Wheeler	UtilityVehicles	Water	Grand Total
1	1	1	cl 11 - ships		0.14%				0.32%	2.59%			94.51%	1.83%
2	2	2	cl 12 - aircrafts		0.27%		98.25%			2.96%		0.47%		2.39%
3	3	3	cl 3 - batteries	1.36%	32.88%	1.12%			2.24%	0.37%	1.20%	0.94%		5.53%
			cl 13 - materials	0.05%	12.88%				0.32%					1.85%
4	4	4	cl 4 - models production	32.39%	3.56%	0.28%		0.38%		8.15%	3.61%	6.35%	1.10%	13.15%
			cl 10 - joint.venture	20.49%	36.44%	1.68%		0.76%	2.24%	2.96%	4.82%	15.53%	2.20%	14.46%
	5	5	cl 6 - trucks & hydrogen	3.15%	10.27%	3.92%	1.75%	13.14%	2.24%	4.07%	6.02%	45.18%	2.20%	8.75%
			cl 7 - services fleets & sharing	3.04%	1.64%	23.67%		27.43%	1.60%	8.89%	45.18%	11.29%		10.27%
	6	6	cl 16 - incidents	0.33%	0.41%	0.14%		0.19%		5.19%				0.48%
		7	cl 8 - court.ruling.lawsuit	1.63%	0.82%	0.28%		3.05%	4.79%	39.63%	7.83%	0.24%		3.66%
	7	8	cl 15 - market	3.64%								0.47%		1.33%
	8	9	cl 5 - subsidy/tax/regulation	2.77%	0.27%	1.82%		9.52%	75.40%	8.52%	2.41%	1.65%		7.44%
5	9	10	cl 2 - electric motor performance	26.74%	0.14%	0.70%		1.71%		2.59%	27.71%	13.41%		11.89%
6	10	11	cl 17 - race	0.98%						0.37%	0.60%			0.39%
		12	cl 14 - motorsport	3.15%		0.14%				4.81%	0.60%	0.24%		1.43%
7	11	13	cl 1 - charging	0.27%	0.27%	63.31%		1.90%	2.88%	6.30%		0.24%		9.56%
8	12	14	cl 9 - buses			2.94%		41.90%	7.99%	2.59%		4.00%		5.59%
Grand Total				100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%

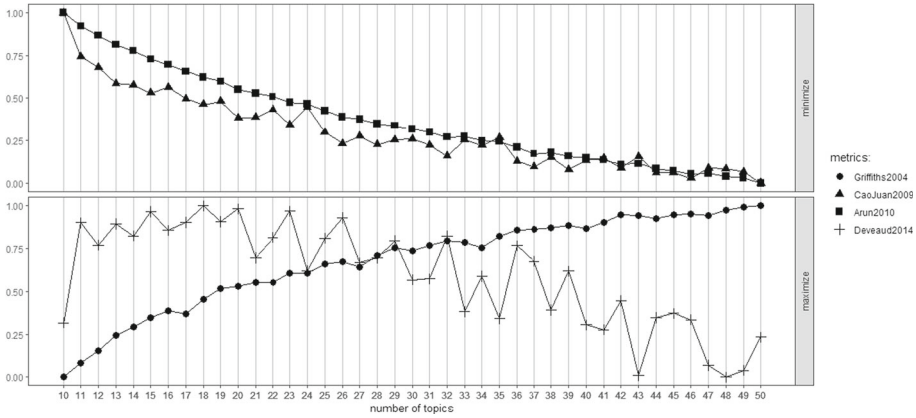

Fig. 4 LDA: optimal number of topics according to Arun, Cao, Deveaud, Griffiths indexes

system. A second group broken down is the large cl-4 (28%) that highlights news items on car models over against the ones on business models in the sustainable mobility sector (joint ventures in particular). A third group broken down accounts for almost 19% of news items dealing with other-than personal car mobility, for example, trucks & hydrogen (cl-6) and service fleets & sharing (cl-7).

4.1.2 LDA

Several methods, based on different metrics, have been put forward to find the optimal number of topics for LDA [11]. As regards the present paper, four specific methods are referred to, namely, Arun [3], Cao [17], Deveaud [23], Griffiths [31], available in R's *ldatuning* library [47]. Figure 4 illustrates how these measurements give different indications. On the one hand, Arun, Cao and Griffiths indices show the best results as the number of topics increases while, on the other, the Deveaud index, based on the maximisation of the information divergence between all pairs of topics, identifies an optimum zone between 18 and 20 topics, consistent with the 17 groups found with the LCA method (see above, in this section).

4.1.3 Factorial models

In the case of factorial methods, the most commonly adopted one consists in selecting a number of dimensions capable of capturing as much variability as present in the original data. In this regard, however, Landauer et al. [42] should be borne in mind:

> "The number of dimensions retained in LSA is an empirical issue. Because the underlying principle is that the original data should not be perfectly regenerated but, rather, an optimal dimensionality should be found that will cause correct induction of underlying relations, the customary factor analytic approach of choosing a dimensionality that most parsimoniously represents the true variance of the original data is not appropriate".

Considerations such as this can also be held valid for other factorial methods, such as NMF.

4.2 Semantic coherence

4.2.1 Benchmark at 17 groups

In assessing the semantic coherence, in this paper the choice of 17 topics refers to the number thereof identified by the LCA method, described above, which show a significant difference in the interpretation of their content by experts. LCA enables the interpretation of a larger number of topics, especially by comparing the possible results with the 10 categories defined by the experts. An economic interpretation of results suggests a cut-off at 17 clusters, which allows for a disaggregation of the expert classification of news items in the vast category "Automobile", but also some significant disaggregation of technical and economic features. For the purpose of identifying the electric mobility eco-system, disentangling those technical and economic domains is an essential contribution in facilitating a socio-technical analysis of the current transition occurring in the mobility sector. Greater details of results obtained with the LCA would not produce significant information for the socio-technical analysis and the number 17 in relation to the quantity of topics becomes the benchmark for all the other methods. Apart from ALC, the other methods under analysis (LDA, LSA, NMF), would not allow a hierarchical branching of results, nor would their optimal number of clusters, being very large, provide a clue for a socio-technical analysis.

4.2.2 Semantic coherence of the 17 topics

The first criterion used to compare topic detection methods is each algorithm's ability to express semantically coherent topics. In this perspective, two coherence indexes are implemented, both referring to 17 topics defined as the benchmark in the present analysis. Box plots in Fig. 5 show the results of measures based on normalised pointwise mutual information, NMPI, (left panel) and UMass score (right panel), respectively. According to NMPI, ranking of relative coherence shows the highest median value for the LCA method, with a high dispersion and cl-8 as outlier (which in fact has a miscellany of issues, "court.ruling.lawsuit"). With UMass score, LDA performs better, both in terms of median and overall topics, while

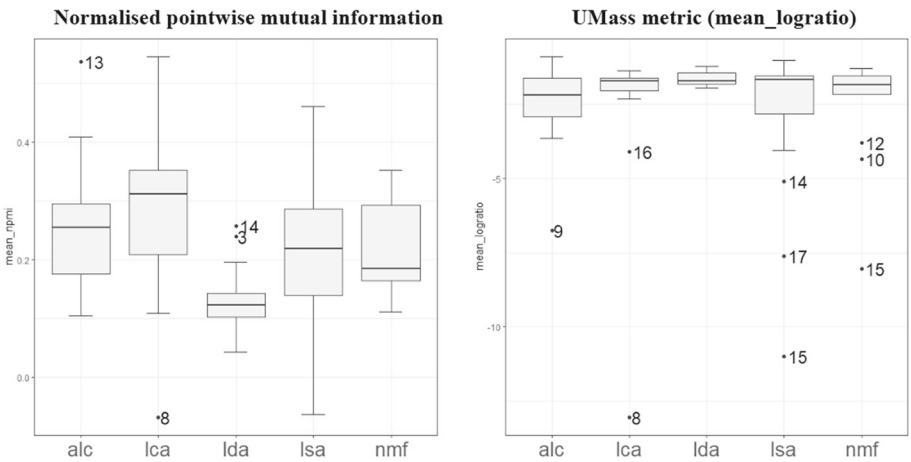

Fig. 5 Box plots of coherence indexes

as regards the LCA method, it highlights not only the case of cl-8, but also of cl-16 (close by in the cluster, for which see the dendrogram in Fig. 1).

4.3 Semantic similarity

A further comparison of the results obtained with the different models focused on the analysis of their semantic similarity. For this purpose, each of the 85 topics (five methods, namely, LDA, LSA, LCA, NMF and ALC, with 17 topics for each method) was associated with a text consisting of the 10 most significant terms. The criteria for selecting terms per topic are were, for LSA and NMF, the terms that most characterise the factorial axes; for LDA the most probable terms for the topic; for LCA the terms with the highest test-value and for ALC the highest chi-square. The total number of different terms under analysis is 282. The matrix on which semantic similarity is assessed is sparsity = 96.3%, but this index does not change substantially by varying the number of terms per topic (with 30 terms it would be 95.9%). At any rate, the texts in question led to the construction of a method/topic-term matrix on which a cosine distance matrix was then calculated (Fig. 6). The hierarchical cluster analysis carried out on the basis of this matrix made it possible to identify 18 clusters, which facilitated comprehension as to which methods and topics are most similar to each other. Figure 7 illustrates the dendrogram of the hierarchical clustering.

An analysis of the silhouette index (Fig. 8) shows some strongly characterised groups. For example, clusters 3 and 9, in which the topics show a high semantic similarity. For these two groups, Table 2 provides the average cosine distance from other topics in the cluster, the silhouette width and the most distinctive terms for each of the six methods.

On the other hand, some groups are certainly less homogeneous, such as cluster 4 or 18 (details in Table 3).

4.4 Classified documents comparison

The heat maps in Annex 3 illustrate the overlap when grouping news items by comparing pairs of analysis methods. The pattern of overlapping across the 15 combinations of pairs of methods is not straightforward. Indeed, for some methods there are more areas of classification that overlap while in others overlapping is less significant. To provide a measure of such overlapping, emphasis is placed on pairs of methods (see results in Table 4), observing a significant association between them (all p-values of the chi-squared test are < 0.001). A summary measure of the strength of the association is provided by Cramer's V. It shows that LSA is the method that most closely approximates expert classification (67–68%), followed by LCA. The other methods under analysis that can be read in terms of hierarchical clustering show that ALC has the lowest Cramer's V, i.e. the lowest overlapping with all the other methods, and also with respect to the expert classification.

5 Discussion

The comparative analysis was based on common ground in the pre-processing of texts, which was then adopted in the implementation of the different methods of topic detection and classification of textual units. A critical issue in the cross method comparison is the choice of the number of topics to be used as a benchmark. In arguing the rationale of this choice, this paper complements techniques for the identification of the optimal number of topics

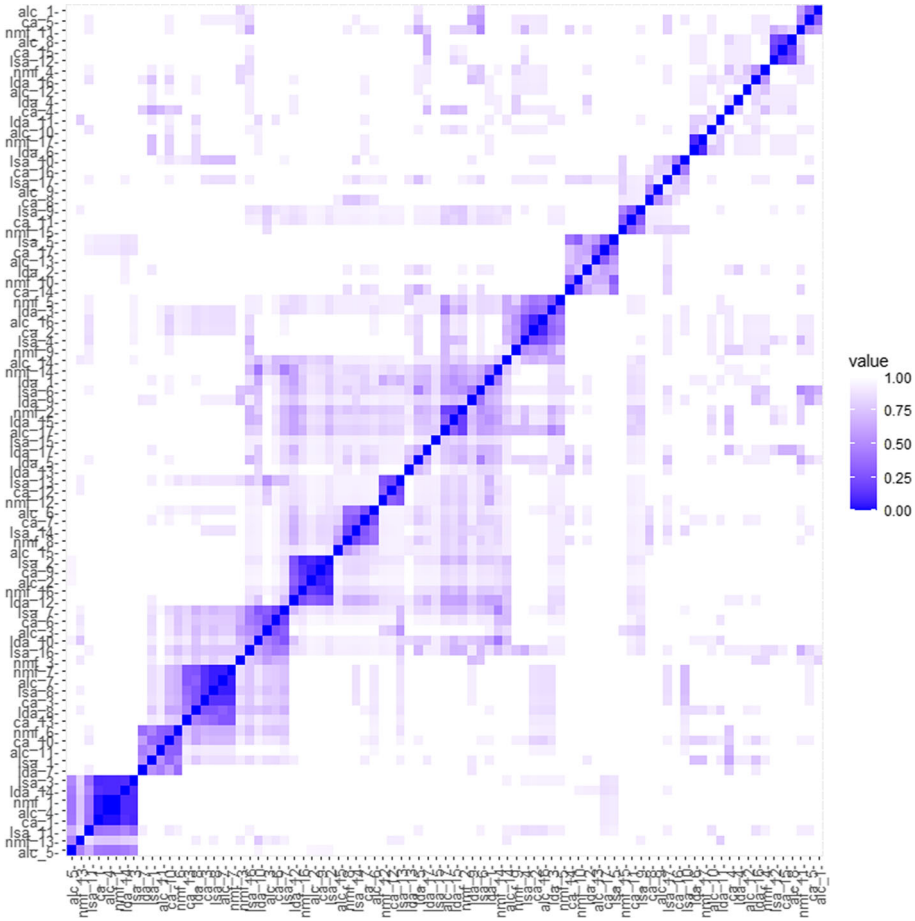

Fig. 6 Distance Matrix method/topic-term of the top ten terms across the 17 topics by each method of topic detection

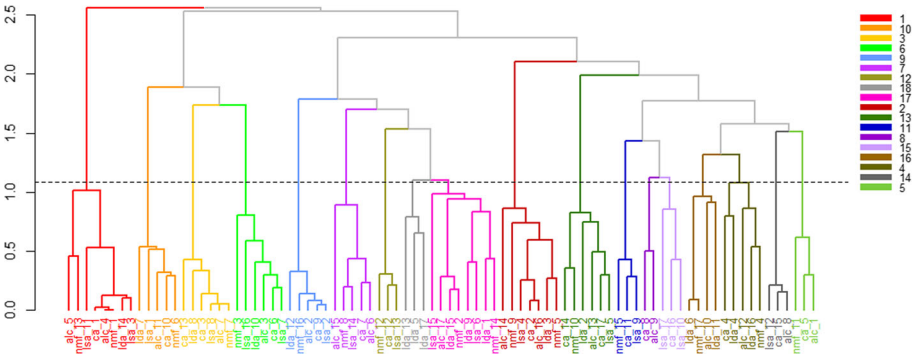

Fig. 7 Cluster analysis dendrogram – method/topic-term matrix

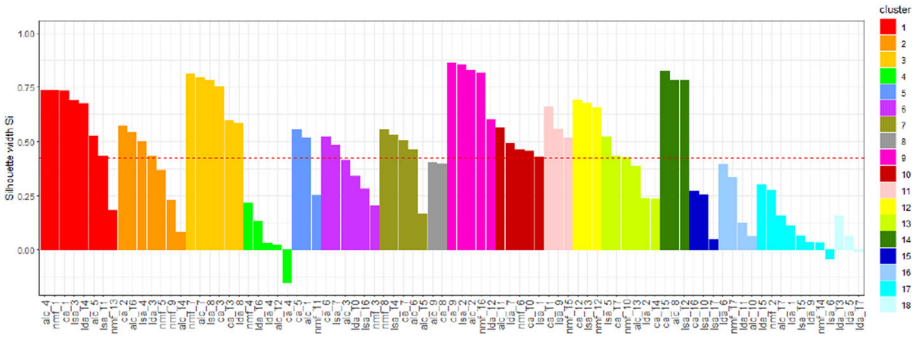

Fig. 8 Silhouette plot – method/topic-term matrix

with the specific research questions that need to be addressed in the identification thereof. The analysis therefore focused on looking not so much at "How many topics are there?", as at "What are the topics that are relevant in the analysis of the socio-technical domains?" Thus, in addition to the identification through the use of given methods, the optimal number of topics needs to be defined according to the interpretive needs of the researcher, who may or may not need more or less detail in the analysis (more or less clusters, for example).

In this paper, methods of topic detection, comprising different characteristics and purposes, have been compared, with the objective of providing an unambiguous classification of documents/news items according to the topics covered. While methods such as LCA and ALC lead to determining the prevailing topic that characterises each document, methods such as LSA, NMF and LDA provide a distribution of importance of different topics in individual documents. From this distribution it is obviously possible to derive the most significant topic, but the possibility that multiple topics characterise a document should be borne in mind. This second group of methods is in fact geared more towards providing an information retrieval tool than a document classification tool. In the comparative analysis, three dimensions have been explored, namely, the semantic coherence of topics identified by the various methods, the semantic similarity of said topics and, finally, the degree of association of elementary units across topics detected by the different methods.

The two measurements of semantic coherence, normalised pointwise mutual information and UMass score show a better result for LCA and for LDA, respectively. With regard to these latter two, better results would have been obtained if a much larger number of clusters had been taken. However, in the comparison it was decided to refer to a given number of cluster in all methods. The measurement of semantic similarity addresses the question of the extent to which different methodologies lead to the formation of homogeneous groups of topics. Results cannot be generalised, since, in some cases, groups of topics are relatively more homogeneous than in others, making the semantic interpretation more difficult. With respect to the strength of the association among topics resulting from the different methods, the Cramer's V shows that LSA is the method that most closely approximates expert classification, followed by LCA. ALC, the other method under analysis that can be read in terms of hierarchical clustering, has the lowest Cramer's V. From the interpretative point of view of the topics encompassed in each cluster, the two methods that offer the hierarchical structure of texts of news items (LCA and ALC) seem to be advantageous, since they enable the definition of a greater or lesser number of topics, thus passing from greater to lesser detail while preserving the hierarchy of topics. However, LCA has a very high overlap with

Table 2 Text comparison. Example of clusters of topics with high semantic similarity, by method of topic detection: average cosine distance from other topics in the cluster (avg.dist) and silhouette width

Cluster	Method_topic	avg.dist	silhouette width	Terms
3	nmf_7	0.1508	0.8119	Cell battery solid.state material lithium recycling cobalt lithium.ion battery.cell process
3	alc_7	0.1622	0.7976	Cell battery lithium material recycling battery_cell cobalt solid-state lithium-ion nickel
3	lsa_8	0.1653	0.7838	Cell solid.state battery battery.cell material lithium recycling cobalt lithium.ion project
3	lca_3	0.2012	0.7533	Cell battery lithium solid.state material battery.cell raw.material recycling research lithium.ion
3	lca_13	0.3472	0.5977	Material nickel battery process recycling cobalt lithium cell electrolyte electrode
3	lda_8	0.3042	0.5852	Battery cell battery.cell process project material research lithium.ion technology company
9	lca_9	0.1199	0.8623	Buses electric.buses e.buses bus depot order ordered city transport.operator bus.fleet
9	lsa_2	0.1127	0.8554	Electric.buses buses city bus e.buses order ordered electric.bus transport.operator depot
9	alc_2	0.1540	0.8279	Electric_buses buses e-buses bus transport_operator depot order bus_fleet ordered articulated_buses
9	nmf_16	0.1414	0.8173	Electric.buses buses bus electric.bus order e.buses ordered city depot train
9	lda_12	0.2650	0.5986	City buses fleet order electric.buses service vehicle electric year bus

all the other methods and appears to be the best method in terms of readability of topics. In addition, when compared with the expert classification, results of LCA (Table 1) allow significant categorisation of domains that enable the identification of the many different sets of agents, the countries in which they are located and policies to enhance production, use and control of artefacts.

6 Conclusion

This paper contributes to the literature on text analysis with a cross method comparison. Following Lebart [43], the comparative analysis to other models focusing on three groups

Table 3 Text comparison. Example of clusters of topics with low semantic similarity, by method of topic detection: average cosine distance from other topics in the cluster (avg.dist) and silhouette widt**h**

Cluster	Method _topic	Avg.dist	sil_ width	Terms
4	nmf_4	0.7342	0.2185	Financing investor quarter employee profit report delivery startup million million.euro
4	lda_16	0.7770	0.1328	Company investment announced plan million million.euro deal year investor billion
4	lda_4	0.8842	0.0308	Year customer announced start end market delivery time update launch
4	alc_12	0.8955	0.0228	Investor shares financing startup company shareholder financial share_price debt takeover
4	ca_4	0.8623	− 0.1501	Model production brand plant platform delivery quarter report sales year
18	lda_13	0.7552	0.1574	New design presented vehicle concept front show feature interior car
18	lda_5	0.6816	0.0639	Electric.car euro price car vehicle subsidy new cost costs make
18	lda_17	0.7254	− 0.0057	Cent year vehicle increase sales quarter electric.car unit sold number

Table 4 Comparison of document classification by pair of methods: chi-squared and Cramer's V

Pair of methods		Chi-squared	p-values	df	Cramer's V
LSA	CATEGORY	21,839.2	0.001	144	0.6822
LCA	CATEGORY	21,486.7	0.001	144	0.6767
LCA	LSA	37,684.4	0.001	256	0.6725
NMF	CATEGORY	17,986.6	0.001	144	0.6187
LCA	NMF	30,067.6	0.001	256	0.6002
LSA	NMF	28,301.4	0.001	256	0.5822
LDA	NMF	27,300.1	0.001	256	0.5717
LDA	CATEGORY	14,096.4	0.001	144	0.5471
LCA	ALC	24,834.6	0.001	256	0.5450
LCA	LDA	24,750.3	0.001	256	0.5441
LDA	LSA	24,184.2	0.001	256	0.5377
ALC	CATEGORY	12,859.6	0.001	144	0.5223
ALC	LDA	22,410.6	0.001	256	0.5174
ALC	LSA	22,271.5	0.001	256	0.5158
ALC	NMF	22,170.8	0.001	256	0.5146

of methods has been extended. Respectively, these are probabilistic methods, factorisation models and clustering models. The five methods selected, LDA, LSA, NMF, LCA and ALC, are compared with an expert classification. The goal was not so much to compare all the possible methods, as to make clear that, given the research questions and the specific features of the corpus under analysis, several different methods can be applied. In particular, the paper contributes by providing tools for cross method comparison, a question left open by Lebart. In this respect, criteria and tools presented herein provide a kit of resources that can support researchers in assessing the results of the various topic detection models that are applied to a given corpus. In addition to different criteria to define the number of topics to be analysed (an optimal number or a predefined number), the cross-method comparison needs to define a given number of topics for comparing methods with respect to the relative semantic coherence of topics that they detect. In turn, the measures of semantic coherence proposed in the paper needs the definition of two parameters, namely, the number of topics and the number of terms per topic. Given the characteristics of the corpus under analysis, neither of these parameters can be set a priori without knowledge of the specific results of topic detection arising therefrom. They must be specified by interpreting the significance of the results obtained. Once defined, both parameters impact on the analysis of the semantic similarity across topics detected with the five methods.

When the researcher decides among methods for identifying topics within the corpus to be analysed, she or he is faced not only with several methods of topic detection, but also with an array of criteria to define the most suitable methods and the relative parameters to decide which method performs better with respect to her or his research questions. In this comparative study, only a few of the most widely-used methods in the literature have been selected. Among them, the most widely-used is undoubtedly the LDA. This paper has focused on the purpose of identification of domains relevant to the interpretation of a new, emerging field in sustainable mobility and the need to classify similar textual units according to different levels of detail within topics. These research questions have guided the choices of comparison between methods. The comparison reveals that, in some respects, the results of the two methods are comparable. However, that LCA provides a hierarchical classification of topics and a unique association of each news item to a dominant topic, which fits the demands of this analysis more effectively, is convincing. These two features have supported the interpretation of the results obtained, but have also opened up the need to examine certain methodological aspects more deeply.

In particular, the cross-method comparison proposed in this analysis refers to the hierarchical clustering associated to the Lexical Correspondence Analysis as a benchmark. This decision builds on the research perspective adopted in the selection of a specific corpus on electric mobility. An alternative research path, to be explored in the future, may look at the cross-method comparison by picking the optimal number of topics to be defined in each method, thus opening up another specific avenue of study that should focus on the various methods used to define the optimal number of topics for each method. In addition, each method needs the setting of alternative parameters at various stages of implementation, meaning that alternative values could be explored to test the robustness of cross-method comparison under a different conditional setting of parameters.

To conclude, it might be argued that the different methods have been applied to a corpus with specific characteristics and that each news item is relatively short, and therefore focuses on one main topic. However, a general issue to be discussed is the specificity of cross-method results with respect to the characteristics of this corpus. It is therefore open to future research to explore the comparison of topic models in cases of corpora with different structural features, in particular with regard to the variety of topics they might include in each document. And

this consideration brings us to a possible conclusion to be kept in mind: there is no one-fits-all method for identifying topics.

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Data availability The data analysed is a collection of English-language news published by electrive.com, more details can be found in Sect. 2.

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

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